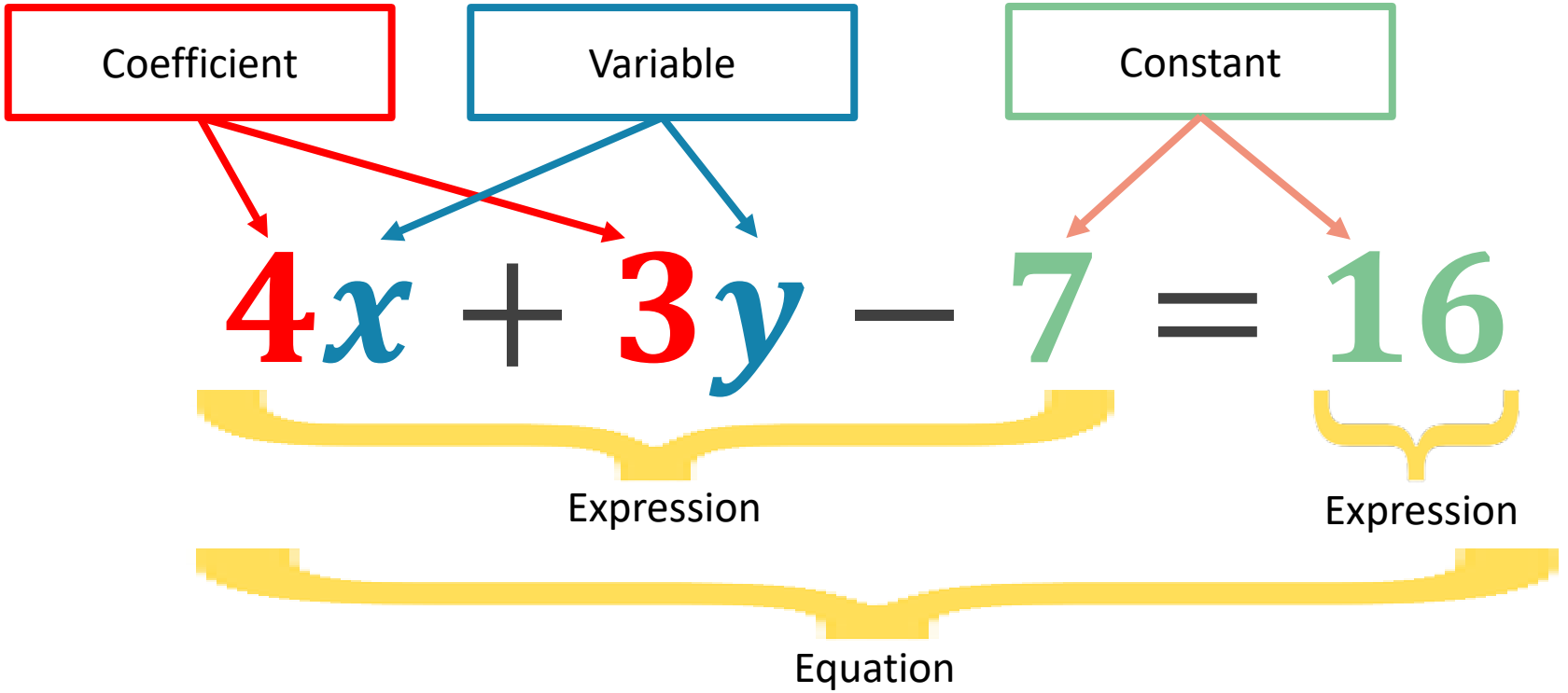
Algebra for Engineering #1



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Algebra terminology



Terms: $4x$, $3y$, -7, 16

Collecting like terms

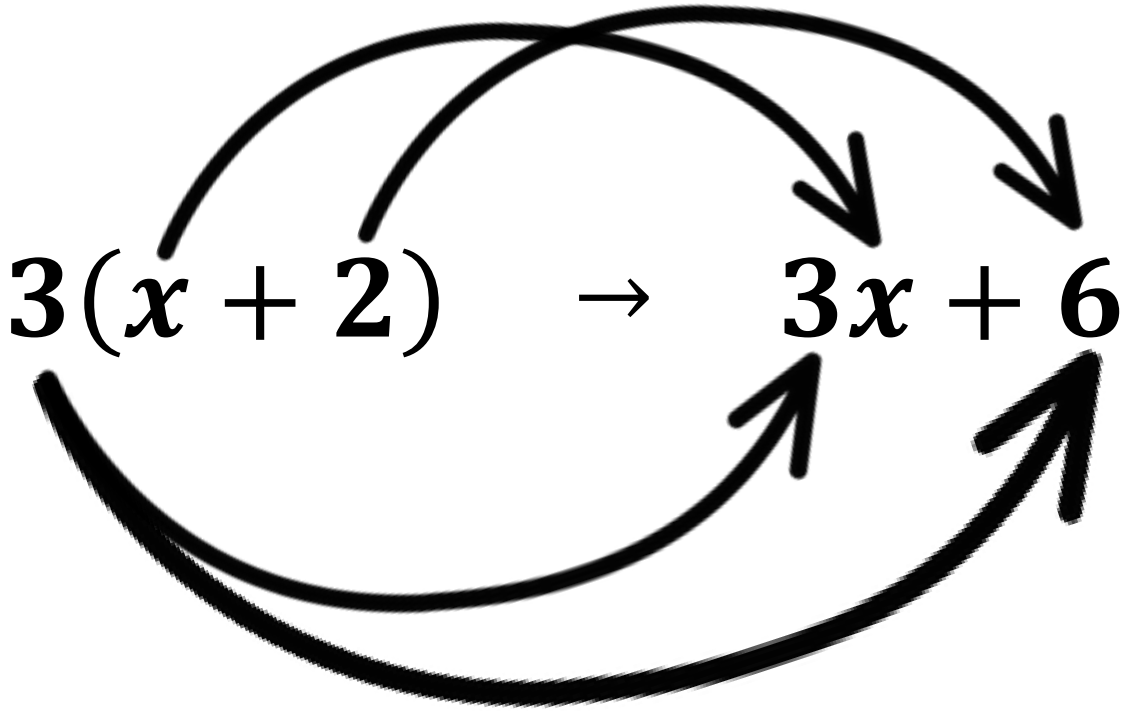
- When there are multiple terms with the same variable in them, we can simplify

$$2x + 5x + 3y = 4 \quad \rightarrow \quad 7x + 3y = 4$$

$$4x + 3 - 2x - 8 \quad \rightarrow \quad 2x - 5$$

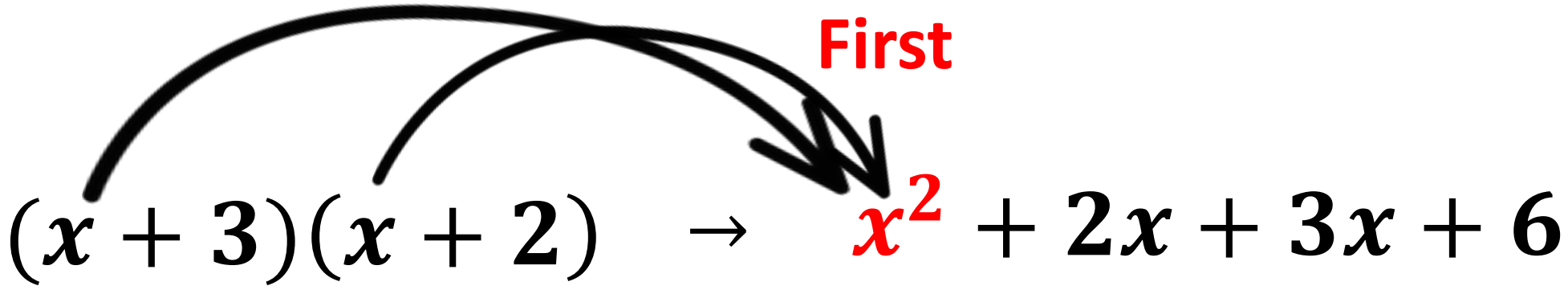
Expanding Brackets

- When working with brackets we have to multiply every term inside the brackets with every term outside the brackets



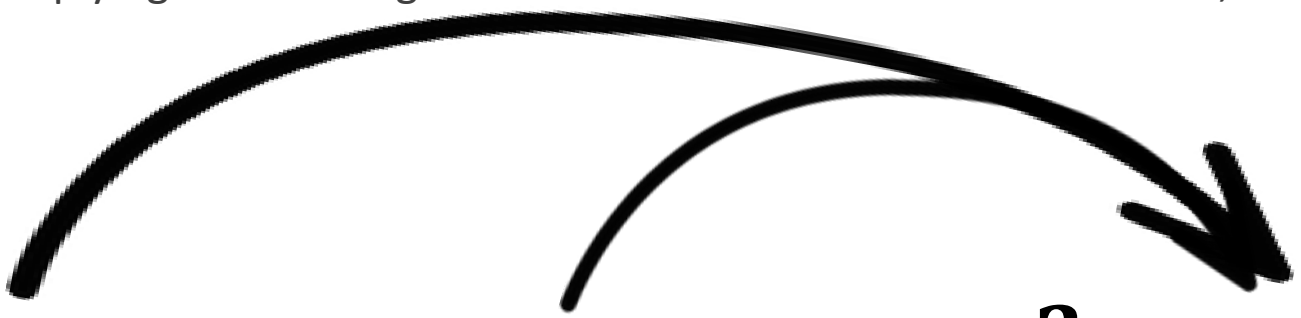
Expanding Multiple Brackets

- When multiplying brackets together we can use the **FOIL** method – First, Outer, Inner, Last



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$$(x+3)(x+2) \rightarrow x^2 + 2x + 3x + 6$$

Expanding Multiple Brackets

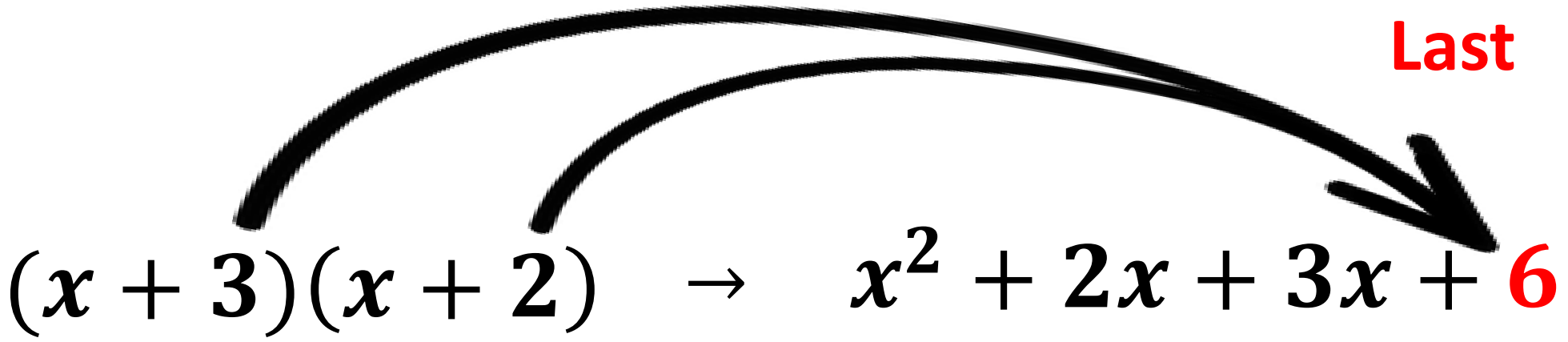
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Expanding Multiple Brackets

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Factorisation

- When factorising equations we need to find the largest common denominator

We know that: $\mathbf{3(x+2)} \rightarrow \mathbf{3x+6}$

Therefore, we also know: $\mathbf{3x+6} \rightarrow \mathbf{3(x+2)}$

3 is the highest common factor here, between both term

Solving Equations

- When solving equations, we need to simplify as much as possible, we also must make sure we are doing the same to both sides of the equation

$$3x + 4 = 19$$

-4 -4

$$3x = 15$$

÷3 ÷3

$$x = 5$$

